
ME 602: Mechanical Vibration

Course Code	ME 602
Course Name	Mechanical Vibration
L-T-P-C	2.5-0.5-0-3
Prerequisites	Consent of the faculty member
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**
Free and forced vibrations with and without damping. [Lectures]
- **Module II**
Vibration isolation and transmissibility; Un-damped vibration absorbers. [Lectures]
- **Module III**
Generalized coordinates and coordinate coupling; Orthogonality of modes. [Lectures]
- **MDOF systems**
Free and forced vibration of multi-degree of freedom systems with and without viscous damping; Lagrange's equation; Holzer's method; Solution of Eigen value problem, transfer matrix and modal analysis. [Lectures]
- **Self excited vibrations**
Criterion of stability; Effect of friction on stability. [Lectures]
- **Continuous Systems**
Vibrations of strings; Free and forced longitudinal vibrations of prismatic bars; Ritz and Galerkin methods. [Lectures]
- **Diagnosis**
Introduction to diagnostic maintenance and signature analysis. [Lectures]
- **Nonlinear Vibration**
Introduction to Nonlinear Vibration. [Lectures]
- **Random Vibration**
Introduction to Random Vibration. [Lectures]
- **Numerical Integration methods in Vibration Analysis**
Finite difference method, Runge-Kutta method, and Newmark method. [Lectures]
- **Finite Element Method**
Equation of motion of an element, Mass matrix, stiffness matrix and Force vector for Bar element, Torsion element and Beam element. Consistent and Lumped mass matrices. [Lectures]

Text and Reference Books

1. S. S. Rao, **Mechanical Vibrations**, Pearson Education Inc. (4th Ed.)2007
2. Leonard Meirovitch, **Fundamental of Vibrations** Mc-Graw Hill Inc.2001
3. D. J. Inman, **Vibration and Control**, John Willey & Sons Inc2002
4. S. Tamadonni & Graham S. KellyM, **Mechanical Vibrations**, Schaum's Out line Series, Mc-Graw Hill Inc 1998